

# CCNA 200-301 Notes

## Chapter 06 - IPv4 Addressing Part 2

### Learning Objectives

Understand Subnet Mask, Private/Public IP, NAT and ARP.

### Subnet Mask

- Determines the Network ID and Host ID.
- Same Network: 192.168.1.5/24 & 192.168.1.100/24.
- Different Network: 192.168.1.5/24 & 192.168.2.5/24.

### Private IP

10.0.0.0/8  
172.16.0.0 - 172.31.255.255  
192.168.0.0/16

### Public IP

Assigned by the ISP. Visible on the Internet. Unique globally.

### NAT

Router translates many Private IPs into one Public IP.

### ARP

If destination MAC is unknown:

1. ARP Request (Broadcast).
2. Destination replies with ARP Reply (Unicast).
3. MAC is stored in the ARP Cache.
4. ICMP Ping is then sent.

### Packet Tracer

Green packets = ARP  
Red packets = ICMP.

### Exam Tips

- Memorize Private IP ranges.
- ARP works only inside the LAN.
- NAT changes IP, not MAC.
- First Octet determines the IP class.